

2-descent when $E[2]$ is generic

The S_3 case: what the full descent computes, why there is no partial descent over $\mathbb{Q}$, and what takes its place

worked examples computed by descent-s3.gp

“Partial 2-descent” has two readings, and in the generic case they have opposite answers. Read as **descent along a 2-isogeny**, it does not exist over $\mathbb{Q}$ at all — Section 6 proves this — and what takes its place is a descent along a factorisation of f over an **extension** (Section 7). Read as **doing only some of the work** — imposing a proper subset of the local conditions — it exists, is cheap, and is what one actually does first (Section 8). Both are treated below.

1. The generic hypothesis, and what it rules out

Write $E : y^2 = f(x)$ with $f = x^3 + c_2x^2 + c_1x + c_0 \in \mathbb{Q}[x]$ squarefree, and let $\Omega = \{\theta_1, \theta_2, \theta_3\}$ be its roots. The 2-torsion is

$$E[2] = \{O, (\theta_1, 0), (\theta_2, 0), (\theta_3, 0)\},$$

so $G_{\mathbb{Q}}$ acts on $E[2] \setminus \{O\}$ exactly as it permutes Ω , and

$$\mathrm{Gal}(\mathbb{Q}(E[2])/\mathbb{Q}) \cong \mathrm{Gal}(f) \subseteq S_3.$$

The **generic** case is $\mathrm{Gal}(f) = S_3$, i.e. f irreducible with disc f not a square.

The structural statement, which everything below uses:

Lemma 1. Let $F_2[\Omega]$ be the permutation module on the roots and $\sigma : F_2[\Omega] \rightarrow F_2$ the sum-of-coordinates map. Then

$$0 \longrightarrow E[2] \longrightarrow F_2[\Omega] \xrightarrow{\sigma} F_2 \longrightarrow 0$$

is exact as $G_{\mathbb{Q}}$ -modules: $E[2]$ is the **sum-zero** submodule of $F_2[\Omega]$.

Proof. The three non-zero points of $E[2]$ sum to O (they are the intersections of E with $y = 0$, a line), so under $E[2] \setminus \{O\} \cong \Omega$ the group law on $E[2]$ is $T_i + T_j = T_k$ for $\{i, j, k\} = \{1, 2, 3\}$, which is the sum-zero subspace of F_2^3 . Galois permutes both sides compatibly. ■

For $\mathrm{Gal}(f) = S_3$ this identifies $E[2]$ with the **standard** 2-dimensional irreducible $F_2[S_3]$ -module, and $G_{\mathbb{Q}} \twoheadrightarrow \mathrm{Aut}(E[2]) = \mathrm{GL}_2(F_2) \cong S_3$ is onto. Three consequences, all immediate and all used later.

Corollary 2. In the generic case:

- (a) $E(\mathbb{Q})[2] = 0$, since $E[2]^{G_{\mathbb{Q}}} = 0$: a fixed non-zero point is a rational root of f .
- (b) $E[2]$ has **no** $G_{\mathbb{Q}}$ -stable line, so **E admits no 2-isogeny over $\mathbb{Q}$** .
- (c) $\mathrm{End}_{G_{\mathbb{Q}}}(E[2]) = F_2$. Indeed the six elements of $\mathrm{GL}_2(F_2)$ span $M_2(F_2)$ over F_2 , so the centraliser of the image is the centre of $M_2(F_2)$.

Part (c) is worth recording because it says the module admits no non-trivial endomorphism to exploit: the rank-one $\varphi \in \mathrm{End}_G(E[\ell])$ that drive the Brauer classes of the survey (§8.1) simply are not there at $\ell = 2$ in the generic case.

The four cases, for orientation. Everything about 2-descent is governed by how f factors over $\mathbb{Q}$, equivalently by the étale algebra $A = \mathbb{Q}[x]/f$.

f over $\mathbb{Q}$	$A = \mathbb{Q}[x]/f$	$\dim E(\mathbb{Q})[2]$	2-isogenies over $\mathbb{Q}$
irreducible, disc non-square	cubic field, non-Galois	0	none — the generic case
irreducible, disc square	cyclic cubic field	0	none
linear $\times$ irred. quadratic	$\mathbb{Q} \times$ quadratic field	1	one
three linear factors	$\mathbb{Q} \times \mathbb{Q} \times \mathbb{Q}$	2	three

Partial (isogeny) descent lives in the bottom two rows only. The top two rows — f irreducible — have no rational 2-torsion at all, hence no 2-isogeny; the generic case is the first of them.

The family of the involution note is entirely generic. For $f = x^3 + a$ one has $\text{disc } f = -27a^2 = -3 \cdot (3a)^2$, which is a square in $\mathbb{Q}$ only if -3 is — never. So $y^2 = x^3 + a$ has $\text{Gal}(f) = S_3$ **whenever a is not a cube**, and `selmer-involution.typ` is a statement about full 2-descent in the generic case throughout. Its 2-division field $\mathbb{Q}(\sqrt[3]{a}, \zeta_3)$ is the S_3 -closure.

2. The cohomology group the descent lives in

Let $K = \mathbb{Q}[x]/f$, a cubic field in the generic case, and θ the image of x .

Proposition 3. There is a canonical isomorphism

$$H^1(\mathbb{Q}, E[2]) \cong \ker\left(N : K^\times / (K^\times)^2 \rightarrow \mathbb{Q}^\times / (\mathbb{Q}^\times)^2\right),$$

and likewise over every $\mathbb{Q}_v$ with K replaced by $K \otimes \mathbb{Q}_v$.

Proof. $F_2[\Omega] = \text{Ind}_{G_K}^{G_{\mathbb{Q}}} \mu_2$, so Shapiro gives $H^1(\mathbb{Q}, F_2[\Omega]) = H^1(K, \mu_2) = K^\times / (K^\times)^2$, and σ induces the norm. Taking cohomology of Lemma 1 and noting that $H^0(\mathbb{Q}, F_2[\Omega]) = F_2 \rightarrow H^0(\mathbb{Q}, F_2) = F_2$ is onto (the all-ones vector maps to 1) gives $0 \rightarrow H^1(\mathbb{Q}, E[2]) \rightarrow K^\times / (K^\times)^2 \rightarrow \mathbb{Q}^\times / (\mathbb{Q}^\times)^2$. ■

So a class in $H^1(\mathbb{Q}, E[2])$ is a square class of the cubic field with square norm. That is the concrete object every 2-descent in this case manipulates.

The **dimensions** of the local pieces come for free from the module, by Lemma B' of `selmer-involution.typ`: writing $L_v \subseteq H^1(\mathbb{Q}_v, E[2])$ for the image of the Kummer map,

$$\dim_{F_2} L_v = \begin{cases} \dim E[2](\mathbb{Q}_v) & v \text{ finite} \\ v \nmid 2 \\ \dim E[2](\mathbb{Q}_2) + 1 & v = 2 \\ \dim E[2](\mathbb{R}) - 1 & v = \infty \end{cases}$$

and $\dim H^1(\mathbb{Q}_v, E[2]) = 2 \dim L_v$, the local condition being maximal isotropic. Since $\dim E[2](\mathbb{Q}_v)$ counts roots of f in $\mathbb{Q}_v$, this is a table:

f over $\mathbb{Q}_v$	$\dim E[2](\mathbb{Q}_v)$	$\dim L_v$ ($v \neq 2, \infty$)	$\dim L_2$
irreducible, or irred. quadratic factor only	0	0	1
one root	1	1	2
three roots	2	2	3

with $\dim L_\infty = 0$ if f has one real root and 1 if it has three. **Every entry depends only on the splitting of f** , never on the curve — which is exactly the point made in the involution note, and the reason two curves sharing $E[2]$ have local conditions of equal size.

3. The descent map

Proposition 4. For $P = (x_0, y_0) \in E(\mathbb{Q})$ set $\delta(P) = x_0 - \theta \in K^\times / (K^\times)^2$. In the generic case this is defined on all of $E(\mathbb{Q}) \setminus \{O\}$, extends by $\delta(O) = 1$, and gives an injection

$$\delta : E(\mathbb{Q})/2E(\mathbb{Q}) \hookrightarrow \ker(N) \subseteq K^\times / (K^\times)^2$$

which is the Kummer map of Section 2.

Three remarks, each of which is a simplification special to the generic case.

(i) **No exceptional values.** $x_0 - \theta$ is a unit of K unless x_0 is a root of f , i.e. unless $P \in E[2]$; and $E(\mathbb{Q})[2] = 0$ by Corollary 2(a). So **the formula never degenerates** and one never needs the usual “if $x_0 = \theta_i$ then use $f'(\theta_i)$ ” clause. In the split case that clause is unavoidable; here it is empty.

(ii) **The norm is automatically a square.** $N_{K/\mathbb{Q}}(x_0 - \theta) = \prod_i (x_0 - \theta_i) = f(x_0) = y_0^2$. So the image lands in $\ker N$ with nothing to check — and conversely the norm condition is the only global constraint visible before local conditions are imposed.

(iii) **It is a homomorphism**, by one line. Let $P_1 + P_2 + P_3 = O$ with the P_i on the line $y = \lambda x + \mu$. Both $f(x) - (\lambda x + \mu)^2$ and $\prod_i (x - x_i)$ are monic cubics with the same roots, hence equal. Evaluating at $x = \theta$ and using $f(\theta) = 0$,

$$\prod_i (\theta - x_i) = -(\lambda\theta + \mu)^2, \quad \text{so} \quad \prod_i \delta(P_i) = \prod_i (x_i - \theta) = (\lambda\theta + \mu)^2,$$

a square. (If the line is vertical, $P_1 + P_2 = O$ and $\delta(P_1)\delta(P_2) = (x_1 - \theta)^2$.)

4. Full 2-descent, as an algorithm

Let S be the set of places consisting of ∞ , the prime 2, and the primes of bad reduction. Write, for a number field K ,

$$K(S, 2) = \left\{ \alpha \in K^\times / (K^\times)^2 : \text{ord}_{\mathfrak{p}}(\alpha) \equiv 0 \pmod{2} \text{ for all } \mathfrak{p} \notin S \right\}.$$

Step 1 (the global bound). $\text{Sel}_2(E) \subseteq K(S, 2) \cap \ker N$.

Step 2 (the size of that). There is an exact sequence

$$1 \longrightarrow \mathcal{O}_{K,S}^\times / \left(\mathcal{O}_{K,S}^\times \right)^2 \longrightarrow K(S, 2) \longrightarrow \text{Cl}_S(K)[2] \longrightarrow 1,$$

whence

$$\dim_{F_2} K(S, 2) = \underbrace{(r_1 + r_2 + \#S_f)}_{\text{S-units}} + \dim_{F_2} \text{Cl}_S(K)[2],$$

with S_f the primes of K above S . (The S -unit rank is $r_1 + r_2 - 1 + \#S_f$; the extra 1 is $-1 \in \mu(K)$.)

Step 3 (the local conditions). For each $v \in S$ compute $L_v \subseteq (K \otimes \mathbb{Q}_v)^\times / \text{squares}$ as the image of $E(\mathbb{Q}_v)/2$ under $x - \theta$, and keep only those global classes landing in L_v for every $v \in S$.

Step 4 (read off). $\dim \text{Sel}_2(E) = \text{rank } E(\mathbb{Q}) + \dim \text{III}(E)[2]$, since $E(\mathbb{Q})[2] = 0$. In particular $\text{rank } E(\mathbb{Q}) \leq \dim \text{Sel}_2(E)$, with equality iff $\text{III}(E)[2] = 0$.

Two things make Step 3 finite and cheap. First, **the conditions outside S are automatic**: at a good $v \nmid 2$ the local condition is the unramified subgroup, and membership in $K(S, 2)$ already forces a class to be unramified there. Second, **one knows in advance how big L_v is** — the table of Section 2 — so when sampling points of $E(\mathbb{Q}_v)$ one knows exactly when to stop.

Where the cost is. Not in the local conditions, which are a bounded number of computations in $(K \otimes \mathbb{Q}_v)^\times$ modulo squares. It is in Step 2: one needs the **class group and unit group of a cubic field** of discriminant comparable to $\text{disc } f$. That is subexponential, and unconditionally certifying it is the expensive part; most implementations compute it under GRH. This is the one structural respect in which the generic case is **harder** than the split case, where $A = \mathbb{Q}^3$ and only $\mathbb{Q}(S, 2)$ — pure factorisation — is needed.

5. Worked examples

Computed by `descent-s3.gp`; full output in `results/descent-s3.txt`. All three curves are generic, have trivial torsion, and admit no rational isogeny of any degree.

curve	f	$\text{disc } K$	$\text{Cl}(K)$	$\dim K(S, 2)$	after N	$\dim \text{Sel}_2$	rank, $\text{III}[2]$
37a1, $N = 37$	$x^3 - 16x + 16$	148	1	6	3	1	1, 0
389a1, $N = 389$	$x^3 + 4x^2 - 32x + 16$	1556	1	6	3	2	2, 0
571a1, $N = 571$	$x^3 - 4x^2 - 14864x - 678064$	-2284	$\mathbb{Z}/2$	6	≤ 3	2	0, $(\mathbb{Z}/2)^2$

In each case $S = \{2, N\}$ and S_f has three elements: 2 is totally ramified in K , and N factors as $\mathfrak{p}\mathfrak{q}^2$. The column “after N ” is the bound **before any local condition** is imposed, and the last two columns show what the local conditions then achieve: $3 \rightarrow 1$, $3 \rightarrow 2$, $3 \rightarrow 2$. The local conditions are doing genuine work in every case.

5.1. 37a1 in full

$E : y^2 + y = x^3 - x$ becomes $Y^2 = X^3 - 16X + 16$ under $X = 4x$, $Y = 8y + 4$. Then $K = \mathbb{Q}[\theta]$ with $\theta^3 = 16\theta - 16$ is totally real of discriminant 148, class number 1, unit rank 2; $S = \{2, 37\}$, and in K

$$2 = \mathfrak{p}_2^3, \quad 37 = \mathfrak{p}_{37}\mathfrak{q}_{37}^2,$$

so $\#S_f = 3$ and $\dim K(S, 2) = (3 + 0) + 3 + 0 = 6$. The six generators $-1, u_1, u_2$ and three S -unit generators have norms, modulo squares,

$$-1, \quad -1, \quad 1, \quad -2, \quad 37, \quad -37,$$

which span all of $\mathbb{Q}(S, 2) = \langle -1, 2, 37 \rangle$; so the norm map is surjective and

$$\dim(K(S, 2) \cap \ker N) = 6 - 3 = 3.$$

The local conditions are $\dim L_2 = 1$, $\dim L_{37} = 1$, $\dim L_\infty = 1$ (three real roots), and they cut the 3 down to

$$\dim \text{Sel}_2(E) = 1.$$

The generator is visible: $P = (0, 0)$ on the original model is $(X, Y) = (0, 4)$, so

$$\delta(P) = 0 - \theta = -\theta,$$

whose norm is 16, a square as it must be, and which is **not** a square in K (the polynomial $T^2 + \theta$ has no root there). Hence $\text{Sel}_2 = \langle -\theta \rangle$, $\text{rank } E(\mathbb{Q}) = 1$, $\text{III}[2] = 0$.

5.2. 571a1, where the descent does not see the rank

Here K has signature $(1, 1)$ and class number 2, so $\text{Cl}_{S(K)}[2]$ contributes a generator beyond the S -units and $\dim K(S, 2) = 2 + 3 + 1 = 6$ again. The bound after the norm condition is ≤ 3 , the local conditions give $\dim \text{Sel}_2 = 2$ — and the rank is 0. The whole of Sel_2 is $\text{III}(E)[2] = (\mathbb{Z}/2)^2$. **2-descent alone cannot decide this curve**; one needs a second descent, or the analytic rank.

6. There is no partial 2-descent over $\mathbb{Q}$

By **partial descent** one normally means descent along an isogeny φ of degree 2, computing $\text{Sel}^\varphi(E)$ and $\text{Sel}^\hat{\varphi}(E')$ separately and combining them. In the generic case this is unavailable, and it is worth being precise about how completely.

Proposition 5. Let $\text{Gal}(f)$ be S_3 or C_3 . Then over $\mathbb{Q}$:

(a) E has no 2-isogeny, since a 2-isogeny is a $G_{\mathbb{Q}}$ -stable line in $E[2]$, i.e. a fixed non-zero point, i.e. a rational root of f (Corollary 2(b)).

(b) Consequently there is no Cassels-type ratio $\#\text{Sel}^\varphi/\#\text{Sel}^\hat{\varphi}$ to compute, no exact sequence $0 \rightarrow E'(\mathbb{Q})[\hat{\varphi}]/\varphi E(\mathbb{Q})[2] \rightarrow \text{Sel}^\varphi \rightarrow \text{Sel}_2 \rightarrow \text{Sel}^\hat{\varphi}$ to interpolate through, and no factorisation of the descent through a smaller field.

(c) More strongly, $E[2]$ is **irreducible** in the S_3 case, so Sel_2 admits no Galois-stable filtration at all: it is not assembled from smaller Selmer groups in any way.

What this costs, concretely. In the split case $f = x(x - e_1)(x - e_2)$ the descent map goes to $(\mathbb{Q}^\times/\text{squares})^2$ and the whole computation is **factorisation of integers**: no class group, no units of a cubic field. The Selmer group of the congruent number curve is computed this way, and Monsky's matrix is precisely the resulting F_2 -linear algebra. In the generic case none of that is available: the arithmetic of a cubic field is **forced**, and it is forced by Proposition 5.

What **does** remain over $\mathbb{Q}$ is not a partial descent but a **deeper** one: second descent, i.e. 4-descent, which refines Sel_2 by asking which of its classes lift to Sel_4 and so detects the elements of $\text{III}[2]$ that stopped the computation for 571a1. That is a different and more expensive technique, not a cheaper one.

7. What takes its place: descent along a factorisation over an extension

The general form of partial descent, as used for hyperelliptic curves (Bruin–Stoll), is: given $y^2 = f(x)$ and a factorisation $f = g \cdot h$ over a field L , the map $P \mapsto (g(x(P)), h(x(P)))$ modulo squares is a **partial descent map**, weaker than the full one but defined over L . Over $\mathbb{Q}$ the generic f has no factorisation. Over an extension it does, and the minimal one is forced.

The minimal field for a partial descent is K itself — the cubic field, of degree 3, and **not Galois**. Over K , $f = (x - \theta)q(x)$ and E_K acquires the rational 2-torsion point $(\theta, 0)$; the quadratic resolvent field $\mathbb{Q}(\sqrt{\text{disc } f})$ does **not** suffice, because over it the Galois group is still $A_3 = C_3$, which acts irreducibly on F_2^2 .

7.1. The isogeny, explicitly

Shift $x \mapsto x + \theta$. Since $f(\theta) = 0$,

$$f(x + \theta) = x^3 + Ax^2 + Bx, \quad A = 3\theta + c_2, \quad B = f'(\theta),$$

so over K the curve is $y^2 = x(x^2 + Ax + B)$ with 2-torsion point $(0, 0)$, and the 2-isogeny φ with that kernel has image

$$E' : Y^2 = X(X^2 - 2AX + (A^2 - 4B)).$$

The quadratic $x^2 + Ax + B$ has roots $\theta_2 - \theta$ and $\theta_3 - \theta$, so

$$A^2 - 4B = (\theta_2 - \theta_3)^2,$$

a square in the Galois closure but not in K . For 37a1, $c_2 = 0$ and $f' = 3x^2 - 16$, giving $A = 3\theta$, $B = 3\theta^2 - 16$ and $A^2 - 4B = 64 - 3\theta^2$.

The φ -descent over K is then the classical one: $\delta_\varphi(P) = x(P)$ and $\delta_{\hat{\varphi}}$ its analogue on E' , both landing in $K(S, 2)$, with

$$\text{rank } E(K) \leq \dim \text{Sel}^\varphi(E_K) + \dim \text{Sel}^{\hat{\varphi}}(E'_K) - 2.$$

7.2. Feeding the answer back to $\mathbb{Q}$

A bound over K is worth having only if it constrains $E(\mathbb{Q})$. It does, and the bookkeeping is representation theory of S_3 . Let L be the Galois closure, $\text{Gal}(L/\mathbb{Q}) = S_3$, and decompose

$$E(L) \otimes \mathbb{Q} \cong m_1 \cdot \mathbf{1} \oplus m_\epsilon \cdot \text{sgn} \oplus m_{\text{std}} \cdot \text{std}.$$

Proposition 6. With $F = \mathbb{Q}(\sqrt{\text{disc } f})$ the quadratic resolvent and E^Δ the quadratic twist of E by disc f ,

$$\text{rank } E(\mathbb{Q}) = m_1, \quad \text{rank } E(F) = m_1 + m_\epsilon = \text{rank } E(\mathbb{Q}) + \text{rank } E^\Delta(\mathbb{Q}),$$

$$\text{rank } E(K) = m_1 + m_{\text{std}} = \text{rank } E(\mathbb{Q}) + m_{\text{std}}, \quad \text{rank } E(L) = m_1 + m_\epsilon + 2m_{\text{std}}.$$

Proof. K is the fixed field of a transposition $\langle \tau \rangle$, of index 3, and $\text{Ind}_{\langle \tau \rangle}^{S_3} \mathbf{1} = \mathbf{1} \oplus \text{std}$; F is the fixed field of A_3 and $\text{Ind}_{A_3}^{S_3} \mathbf{1} = \mathbf{1} \oplus \text{sgn}$. Now $\text{rank } E(M) = \dim \text{Hom}_{S_3}(\text{Ind}_H^{S_3} \mathbf{1}, E(L) \otimes \mathbb{Q})$ for $M = L^H$. ■

So $\text{rank } E(\mathbb{Q}) \leq \text{rank } E(K)$, and a partial descent over K bounds the rank over $\mathbb{Q}$ — but **with a defect of m_{std}** , which the method does not see. The two-place accounting is:

	full 2-descent over $\mathbb{Q}$	partial (φ -)descent over K
computes	$\text{Sel}_2(E/\mathbb{Q})$ exactly	$\text{Sel}^\varphi, \text{Sel}^{\hat{\varphi}}$ over K
bounds	$\text{rank } E(\mathbb{Q})$	$\text{rank } E(K) = \text{rank } E(\mathbb{Q}) + m_{\text{std}}$
needs	$\text{Cl}(K), \mathcal{O}_K^\times$, and $\ker N$	$\text{Cl}(K), \mathcal{O}_K^\times$ — twice
local work	images of $E(\mathbb{Q}_v)/2$	local indices $\#(E'(K_w)/\varphi E(K_w))$
defect	$\dim \text{III}(E/\mathbb{Q})[2]$	$\dim \text{III}(E/K)[\varphi]$ and m_{std}

Verdict. Over $\mathbb{Q}$ the partial descent is not merely unavailable but not worth simulating: it needs the same cubic field, run twice, and answers a weaker question. Its real use is when the base field **already contains a root of f** — descending over K , or in a tower, or over a function field where the split case is arranged by construction. The one genuine gain is that its local conditions are index computations of Tamagawa type rather than images of Kummer maps.

8. The other reading: doing only part of the local work

The second sense of “partial” is real, cheap, and is what one does first. Each of the following is an upper bound for $\dim \text{Sel}_2$, obtained by **discarding** conditions, and they are nested.

bound	what is used	37a1
$\dim K(S, 2)$	ramification only	6
$\dim(K(S, 2) \cap \ker N)$	+ the norm condition	3
+ the condition at ∞ and at 2	+ two local images	≤ 3
$\dim \text{Sel}_2$	+ all $v \in S$	1

Three remarks on using them.

(i) **The norm condition is free** and is the single most effective cut: in all three examples it removed 3 of the 6 dimensions. It requires no local computation at all, only the norms of the S -unit generators.

(ii) **The parity is free too.** By the p -parity theorem the parity of $\dim \text{Sel}_{2^\infty}$ is the global root number, a product of local root numbers computable from reduction data. In the generic case with $E(\mathbb{Q})[2] = 0$ this pins $\dim \text{Sel}_2$ modulo 2 whenever $\text{III}[2^\infty]$ contributes evenly — and it is often enough to decide between the two remaining possibilities after the cheap bounds.

(iii) **Order the places by how much they cut.** $\dim L_v$ is known in advance from the table of Section 2, so one can predict which places are most restrictive — the ones where f is irreducible over $\mathbb{Q}_v$, forcing $\dim L_v = 0$ (or 1 at $v = 2$) — and impose those first.

9. Relation to the other notes

`selmer-involution.typ` is this machinery applied to $f = x^3 + a$, which by Section 1 is generic for every non-cube a . Its Lemma B' is the dimension table of Section 2; its algebra A_a is the cubic field K of Section 2; its descent map is the $x - \theta$ of Section 3; and its Proposition E — the valuation argument at the primes dividing a — is precisely the computation of L_v that Step 3 of Section 4 calls for. Its main theorem is a statement that two **different** curves give the **same** subspace at every place; the present note is the same computation done once, for one curve, to the end.

`selmer-local-conditions.typ` places all of this in the general Selmer-structure formalism: the algorithm of Section 4 is (F1) finiteness, the table of Section 2 is (F2) self-duality, and the absence of partial descent (Section 6) is the failure of the module to decompose — which is also why the statistics of Sel_2 in the generic case are the cleanest, being the $\text{Gal} = S_3$ case where the Klagsbrun–Mazur–Rubin disparity depends on a single parameter.

Summary. When $E[2]$ is generic there is exactly one descent available over $\mathbb{Q}$, and it is the full one. It computes a subgroup of the square classes of a cubic field with square norm, cut out by one local condition per bad place, each of predictable dimension. It costs the class group and units of that cubic field, and it fails to give the rank exactly by $\dim \text{III}[2]$. Partial descent in the isogeny sense requires a rational 2-torsion point and therefore requires base change to the cubic field, where it answers a weaker question at comparable cost; partial descent in the sense of discarding local conditions is free, and the norm condition alone is usually half the answer.